

Questions9 of 10 points

✓

Why might being required to change our passwords regularly actually pose a threat to our security?

1/1

Changing passwords too often can make people create weak and predictable passwords. Some people also write their passwords down because they cannot remember them. This can reduce security.

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If I have a six-character password consisting of uppercase (English) letters and (decimal) digits only, how many seconds might it take an adversary to crack, assuming they make one attempt per second?

1/1

2176782336✓

✓

Humor us for a moment, and play [The Password Game](#), trying to get through at least Rule 12.

1/1

While obviously the game itself is in many ways meant to be humorous, it also critiques the experience many of us have setting up new passwords. Explain how there's a trade-off between usability and security in the context of passwords.

The password game shows that very strict password rules can make passwords more secure, but they also make them harder to create and remember. A good password system should balance security and usability.

Consider the below comic for the next two questions.



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Consider the top row of the comic above. Why are passwords like those easy (for a computer) to guess but hard (for a human) to remember?

1/1

These passwords are based on common words with predictable substitutions, like replacing letters with number or symbols. Computers can guess these patterns easily, but humans may find them difficult to remember because they are not natural words.
EX - Tr0ub4dor&3

✓

Now consider the bottom row of the comic above. Why are passwords like those hard (for a computer) to guess but easy (for a human) to remember?

1/1

These passwords use random words together, creating many possible combinations. This makes them difficult for computers to guess. At the same time, humans can remember them more easily because they form a simple and meaningful phrase.
EX - correct, battery, cat etc.

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What is a "credential stuffing" attack?

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A credential stuffing attack is when attackers use usernames and passwords stolen from one website to try logging into other websites, hoping that people have reused the same password.

✓

Provide a specific example of something that would be considered a **type of knowledge factor** for authentication purposes.

1/1

Do NOT provide any of your own knowledge factors (or anything resembling them) themselves as an answer to this question. We are looking for you to answer the question in the general sense (a "type of").

If you provide an answer that is, or appears to be, an actual specific knowledge factor, the answer WILL be marked incorrect, without exception, and you should consider that knowledge factor to have been compromised.

a password or passphrase.✗

✓

Provide a specific example of something that would be considered a **type of inference factor** for authentication purposes.

1/1

A fingerprint✗

✓

Why are phishing attacks so difficult to prevent?

1/1

phishing attacks are difficult to prevent because they trick users into trusting fake emails, messages, or websites, even careful people can make mistakes by greed.

✗

Suppose that your boss asks you whether the company should require use of password managers for all employees.

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Explain in a short paragraph why you might want everyone in the company to use a password manager.

A password manager helps employees create and store strong, unique passwords for every account. This reduces password reuse and makes company accounts more secure. It also saves time because users do not have to remember every password.